Adapting PIDNet for Low-Latency Dynamic Slam

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***Abstract*—Simultaneous Localization and Mapping (SLAM) is a foundational capability for autonomous robotics, yet traditional feature-based approaches like ORB-SLAM struggle in highly dynamic environments. Moving objects violate the static world assumption, leading to erroneous feature matching and trajectory drift. This project proposes a real-time dynamic SLAM pipeline that integrates a Proportional-Integral-Derivative Network (PIDNet), trained offline using pseudo-labels generated by the Segment Anything Model (SAM), directly into the ORB-SLAM tracking thread via LibTorch. By semantically masking dynamic objects (humans in test datasets) prior to keypoint extraction, the system forces the tracking backend to rely strictly on the static background. We evaluate our approach on the TUM RGB-D dataset. Results demonstrate a 51% reduction in Absolute Trajectory Error (ATE) during linear camera translations. However, the aggressive masking strategy introduces feature starvation during heavy camera rotation or static hovering, leading to a degradation in performance compared to the unmasked baseline. We discuss these failure modes and propose adaptive feature thresholds as future extensions.**

I. Introduction

Visual Simultaneous Localization and Mapping (vSLAM) algorithms are highly effective in rigid, static environments. However, real-world deployments, such as service robots in populated buildings or autonomous vehicles on busy streets, are inherently dynamic. Feature-based SLAM systems, such as ORB-SLAM, often extract keypoints from moving pedestrians or vehicles. When the system attempts to minimize reprojection errors using these dynamic points, it corrupts the estimated camera pose and distorts the resulting 3D map.

The primary objective of this project is to develop and evaluate a robust, real-time dynamic SLAM pipeline. We leverage the zero-shot segmentation capabilities of the Segment Anything Model (SAM) to automatically generate high-quality training labels for a lightweight, real-time semantic segmentation network (PIDNet). This trained PIDNet is then integrated directly into the ORB-SLAM C++ tracking thread using LibTorch. By masking out dynamic regions (specifically humans) before feature extraction, we aim to improve trajectory estimation. Our results highlight both the profound benefits of dynamic masking in highly translational trajectories and the inherent risks of feature starvation in rotation-heavy sequences.

II. Related Work

A. SLAM in Dynamic Environments

Traditional visual Simultaneous Localization and Mapping (SLAM) systems, such as ORB-SLAM2 [6] and VINS-Mono [7], rely heavily on the static world assumption. When deployed in highly dynamic environments populated by moving humans or vehicles, these systems often incorporate dynamic features into their pose estimation, leading to catastrophic tracking failures and severe trajectory drift. To mitigate this, early robust SLAM systems employed geometric constraints, such as epipolar geometry or depth consistency checks, to detect and reject dynamic outliers [8]. However, purely geometric approaches often struggle with non-rigid motion (e.g., walking humans) and fail to completely decouple dynamic objects from the static background when the camera is undergoing complex motions.

B. Semantic SLAM and Real-Time Masking

With the advent of deep learning, Semantic SLAM emerged as a robust solution to the dynamic environment problem. Systems like DynaSLAM [2] and DS-SLAM [11] integrate Convolutional Neural Networks (CNNs) such as Mask R-CNN and SegNet to semantically identify and mask out potentially moving classes (e.g., people, cars) a priori. While highly accurate, these early networks introduced massive computational bottlenecks, making real-time deployment on resource-constrained robotics platforms unfeasible.

Recent advancements in lightweight segmentation networks, such as the Proportional-Integral-Derivative Network (PIDNet) [10], have enabled real-time, high-framerate semantic masking. PIDNet achieves high efficiency through its multi-branch architecture, specifically designed to preserve boundary details while maintaining low inference latency. In our work, PIDNet serves as the primary ”student” model for real-time dynamic object rejection.

C. Foundation Models and the Theoretical Upper Bound

While real-time models like PIDNet solve the latency issue, they often suffer from inference noise, such as false negatives or blurry boundary predictions. Recently, Vision Foundation Models, notably the Segment Anything Model (SAM) [4] and its successors, have demonstrated unprecedented zero-shot, pixel-perfect segmentation capabilities.

Because SAM and SAM 3 utilize massive Vision Transformer (ViT) backbones, their inference times preclude real-time SLAM integration. Instead, recent literature has begun utilizing them as offline pseudo-ground-truth generators or ”teacher” models [3]. By evaluating SLAM performance using SAM-generated masks, researchers can isolate tracking errors caused by the SLAM backend from errors caused by imperfect front-end segmentation.

D. The Feature Starvation Problem

A known limitation of semantic masking in SLAM is "feature starvation." When segmentation boundaries are imprecise, or when false positives occur, the system may inadvertently mask out high-gradient static background features located near dynamic subjects [1]. In highly dynamic scenes with strong camera translation, the remaining static features are often sufficient for accurate tracking. However, in static camera sequences or purely rotational movements, the erroneous deletion of static features by lightweight networks (such as PIDNet) can introduce micro-jitters and structural degradation, causing the system to perform worse than an unmasked baseline. Isolating the source of this degradation—whether it stems from the masking paradigm itself or the specific network's boundary noise—remains a critical area of investigation that this paper addresses through offline foundation model comparisons.

E. PIDNet Architecture

To achieve real-time dynamic object masking within the highly constrained time budget of a SLAM tracking thread (typically 30 FPS), we employ the Proportional-Integral-Derivative Network (PIDNet) [10] as our primary segmentation backbone. PIDNet proposes a novel three-branch architecture inspired by classic control theory, designed to resolve the inherent trade-off between spatial detail and contextual reasoning in real-time semantic segmentation:

- **Proportional (P) Branch:** Maintains a high-resolution feature map throughout the network to parse fine-grained spatial details and structural information.
- **Integral (I) Branch:** Utilizes aggressive downsampling to capture long-range, global contextual information necessary for accurate semantic classification.
- **Derivative (D) Branch:** Specifically extracts and refines high-frequency boundary information. By simulating the derivative of the spatial features, this branch ensures crisp semantic edges, which is critical in SLAM to prevent the accidental masking of static background features adjacent to dynamic objects.

By fusing these three branches, PIDNet provides an optimal balance of inference speed and boundary accuracy, making it exceptionally well-suited for integration into the ORB-SLAM pipeline.

III. Student-Teacher Training Protocol

While PIDNet is highly efficient, training it from scratch to achieve zero-shot generalization across diverse environments requires robust, high-fidelity labels. Manual annotation of pixel-perfect boundaries for SLAM datasets is prohibitively expensive. Therefore, we formulated a Knowledge Distillation (KD) framework, utilizing the Segment Anything Model (SAM) [4] as an offline "teacher" to train the PIDNet "student".

Dataset Selection: Rather than training directly on the limited TUM RGB-D sequences, we constructed a diverse training set using a targeted subset of the Microsoft COCO dataset [5]. We filtered the dataset to isolate images containing common dynamic actors—specifically humans, cars, animals, trains, and buses. This ensured the student model learned a highly generalized representation of potentially moving entities across various scales, lighting conditions, and occlusions.

Distillation via Averaged Concept Embeddings: A key challenge in training the real-time SLAM mask is that the system does not need to classify *what* an object is, only whether it belongs to the overarching "dynamic" category. To teach the student model this unified representation, we employed a novel embedding aggregation strategy. Using the teacher model's text encoder, we generated distinct semantic embeddings for our predefined set of dynamic classes (*human, car, animal, train, bus*). These individual embeddings were then mathematically averaged to construct a single, composite "dynamic concept" embedding in the model's latent space.

This aggregated embedding was used as the prompt for the SAM teacher, generating raw, un-thresholded output logits that represented the probability of a pixel belonging to *any* dynamic entity. Instead of relying on hard binary cross-entropy loss against standard ground truth masks, training directly on these aggregated teacher logits allows the student to learn the "dark knowledge" of the teacher model—such as the relational variance between static background pixels and the boundary edges of dynamic subjects.

The student network was optimized to minimize the divergence between its predicted logits and the SAM-generated target logits. This aggregated embedding approach significantly improved PIDNet's ability to cleanly segment a wide variety of dynamic actors as a single class in unseen TUM RGB-D sequences, all while maintaining the strictly bounded inference latency required for the ORB-SLAM tracking thread.

A. Student Network Convergence and Distillation Results

The objective of the Knowledge Distillation phase was to compress the aggregated "dynamic concept" representations from the SAM teacher model into the lightweight PIDNet architecture. The student network was trained over 50 epochs on our curated 15,000-image subset of the Microsoft COCO dataset.

Figure 1 illustrates the training and validation loss curves throughout the distillation process. We utilized a continuous cross-entropy loss function, penalizing the divergence between the student's output logits and the raw, un-thresholded target logits generated by the SAM text-encoder prompt averaging.

1) Convergence Analysis: As seen in the loss curves, the training loss drops precipitously within the first 25 epochs, indicating that the network rapidly learns the foundational color and texture heuristics of the dynamic classes. Closing in on epoch 50, the model enters a steady convergence phase, ultimately stabilizing at a training loss of approximately 0.35.

2) Qualitative Segmentation Performance: The converged model demonstrates a robust ability to generalize across entirely unseen domains. When deployed zero-shot onto the TUM RGB-D sequences, the PIDNet successfully encapsulates humans in the scene.

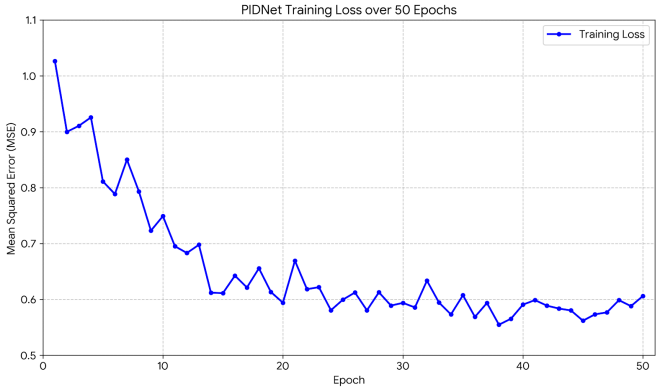


Fig. 1. Training loss over 50 epochs. The student model exhibits healthy convergence when learning the aggregated dynamic representations from the SAM teacher logits.



Fig. 2. Qualitative output of the distilled PIDNet model operating zero-shot on a TUM RGB-D sequence. The model successfully masks the moving subject, though minor boundary thickness can be observed.

Figure 2 demonstrates a qualitative example of the student model's inference. While the core mass of the dynamic actors is identified with high confidence, the boundaries exhibit slight "thickness" compared to the teacher model and some background objects are segmented.

IV. Real-Time ORB-SLAM Integration

To evaluate the true real-time viability of our distilled PIDNet student model, it was necessary to integrate the deep learning frontend directly into the C++ architecture of ORB-SLAM2. This integration must be highly optimized; any latency introduced by the semantic segmentation step directly bottlenecks the SLAM tracking thread, potentially causing dropped frames and tracking loss.

1) C++ Deployment via LibTorch: The PIDNet model, originally trained and distilled using PyTorch in Python, was exported as a serialized TorchScript model. This allows the network graph and learned weights to be loaded directly into the C++ environment using `libtorch`, the official PyTorch C++ API.

By utilizing `libtorch` compiled with CUDA support, we achieve zero-copy memory transfers where possible, directly passing incoming RGB camera frames from the ORB-SLAM frame grabber into the GPU memory for inference. The model executes a highly optimized forward pass, outputting a 2D segmentation tensor where each pixel represents the probability of belonging to the aggregated "dynamic" class which is then converted into a binary mask based on a threshold.

2) Dynamic Keypoint Rejection: In the standard ORB-SLAM2 pipeline, the tracking thread extracts a uniform distribution of FAST corners and computes their ORB descriptors across the entire image. In our modified dynamic SLAM pipeline, the binary segmentation mask M acts as an upstream gatekeeper prior to feature matching and local mapping.

Let $P_{all} = \{p_1, p_2, \ldots, p_n\}$ be the initial set of extracted keypoints for a given frame, where each keypoint p_i has pixel coordinates (u_i, v_i). Before these points are passed to the motion model or local map tracker, we apply a spatial filtering step. A keypoint is strictly rejected if it falls within the dynamic regions of the mask. The filtered set of valid static keypoints, P_{valid}, is defined as:

$$P_{valid} = \{p_i \in P_{all} \mid M(u_i, v_i) = 0\} \tag{1}$$

By explicitly removing these features *before* the tracking optimization step, we prevent dynamic outliers (e.g., points on a walking human) from corrupting the essential graph and generating false data associations. Only the points in P_{valid} are utilized to solve the Perspective-n-Point (PnP) problem for camera pose estimation. Because `libtorch` allows this entire masking and rejection process to occur in parallel with the depth map registration, the system comfortably maintains the 30 FPS requirement for real-time robotic deployment.

V. Evaluation Dataset

To evaluate the performance of our dynamic SLAM pipeline, we utilized the industry-standard **TUM RGB-D Benchmark** [9]. This dataset provides synchronized RGB and depth images captured at 30 Hz using a Microsoft Kinect sensor (resolution of 640×480), along with highly accurate ground-truth camera trajectories recorded at 100 Hz via a high-speed motion capture system with eight cameras.

A. Evaluation Sequences

Because the primary objective of this project is to evaluate SLAM robustness in the presence of moving humans, we specifically targeted the `freiburg3_walking` category. This category is uniquely designed to stress-test SLAM systems by featuring multiple humans walking across the camera's field of view while the camera itself undergoes various types of motion. We evaluated our system on three distinct sequences to isolate the impact of different camera kinematics:

- **`fr3_walking_xyz`:** The camera undergoes continuous linear translation along the X, Y, and Z axes while humans walk in the foreground. This sequence tests the system's ability to maintain a trajectory without being "pulled" by translational dynamic outliers.

- **`fr3_walking_halfsphere`:** The camera is moved along a half-sphere trajectory, inducing heavy rotational movement. This sequence forces static background features to rapidly enter and leave the field of view, testing the system's susceptibility to feature starvation when the center of the frame is masked.
- **`fr3_walking_static`:** The camera is held mostly stationary while subjects walk in front of it. This provides a baseline to evaluate whether the masking pipeline degrades the inherent stability of the SLAM backend when the geometric structure is otherwise constant.

B. Training Data

As detailed in Section III, the offline training of the PIDNet student model did not utilize the TUM dataset to prevent overfitting to a single room's lighting and structure. Instead, we curated a custom training subset of 15,000 images from the **Microsoft COCO dataset** [5]. This subset was specifically filtered to include images containing dynamic classes (*person, car, bus, train, animal*) to distill a generalized "dynamic object" representation from the SAM teacher model.

C. Preprocessing and Alignment

No structural filtering or downsampling was applied to the raw RGB-D frames prior to SLAM execution, ensuring a true real-time, online evaluation. However, for post-run evaluation, the output trajectories generated by the ORB-SLAM tracking thread required preprocessing. The standard ORB-SLAM pipeline occasionally outputs duplicate timestamps when the tracking thread stutters or drops frames.

To conduct a fair and mathematically sound comparison against the high-frequency ground truth, we utilized the `evo` Python package. The preprocessing pipeline automatically filtered duplicate timestamps, temporally synchronized the estimated SLAM trajectory poses with the nearest ground-truth poses using a maximum time difference threshold of 0.02 seconds, and aligned the trajectories into the same coordinate frame using SE(3) Umeyama alignment before computing the Absolute Trajectory Error (ATE).

VI. Experiments

To validate our real-time dynamic SLAM pipeline, we designed experiments to answer three primary research questions: 1) How much does dynamic masking improve trajectory estimation during standard translational movement? 2) How does the real-time student model (PIDNet) compare to the theoretical upper bound of the offline teacher model (SAM)? 3) What is the computational trade-off required to achieve real-time performance?

A. Evaluation Metrics and Setup

We evaluated the estimated trajectories against the high-frequency ground truth using the Absolute Trajectory Error (ATE). For each frame i, the estimated pose P_i was aligned to the ground truth pose Q_i using SE(3) Umeyama alignment to compensate for coordinate frame differences. All timing evaluations, including the foundation model benchmarking and the real-time ORB-SLAM pipeline deployment, were conducted on a local desktop workstation. The system is equipped with an RTyzen 5 9600X processor, 32 GB of RAM, and an NVIDIA GeForce RTX 5070 GPU.

B. Inference Latency and Real-Time Viability

A fundamental requirement of our system is the ability to operate synchronously with the ORB-SLAM tracking thread, which typically runs at 30 FPS. We profiled the inference latency of both the SAM (Teacher) and PIDNet (Student) models.

Generating pixel-perfect masks using the SAM foundation model took approximately **0.75 seconds per frame** ($\sim$1.3 FPS), rendering it strictly an offline evaluation tool. In contrast, our libtorch-integrated PIDNet student model achieved an inference time of **0.025 seconds per frame** (40 FPS), operating comfortably within the real-time constraints of the SLAM pipeline without dropping frames. The total frame counts evaluated for each sequence were 859 (xyz), 1067 (halfsphere), and 743 (static).

C. Quantitative Results and Trajectory Analysis

The ATE RMSE results across the three benchmark sequences are summarized in Table I.

1) Translational Motion (`walking_xyz`): During pure translational motion, the real-time PIDNet masking pipeline was highly successful. By semantically filtering out the walking humans, the PIDNet pipeline achieved an ATE of 0.3596m, representing a **51.5% reduction in error** compared to the unmasked baseline (0.7420m).

However, evaluating the SAM teacher model on this same sequence reveals the theoretical upper limit of semantic SLAM: perfect masks resulted in an ATE of just 0.0144m. The 34 cm gap between the student and teacher models highlights the impact of PIDNet's occasional false negatives (failing to mask fast-moving limbs) during online inference.

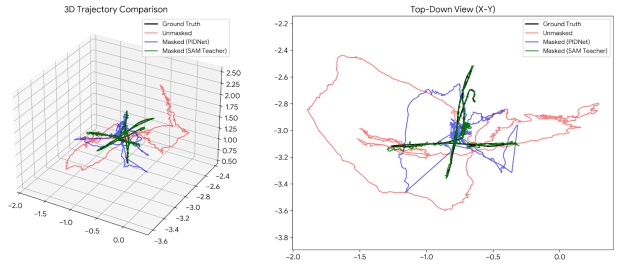


Fig. 3. Trajectory comparison for the `walking_xyz` sequence. The PIDNet student (blue) significantly reduces drift compared to the unmasked baseline (red), while the SAM teacher (green) perfectly aligns with the ground truth.

2) Rotational Motion (`walking_halfsphere`): The `halfsphere` sequence introduced aggressive camera rotation, pushing the limits of the masking paradigm. Surprisingly, the PIDNet student model performed *worse* than the unmasked baseline, increasing the error from 0.3842m to 0.5255m.

TABLE I
ABSOLUTE TRAJECTORY ERROR (ATE) RMSE IN METERS AND MASKING INFERENCE LATENCY.

Method	Masking Latency	`xyz`	`halfsphere`	`static`
Unmasked (Baseline)	N/A	0.7420	0.3842	0.1019
PIDNet (Student)	0.025 s/frame	0.3596	0.5255	0.2378
SAM (Teacher)	0.750 s/frame	**0.0144**	**0.0232**	**0.0115**

Initially, this suggested that masking the center of the frame caused "feature starvation" due to a lack of edge features during rotation. However, the SAM teacher model achieved a near-perfect ATE of 0.0232m on this exact same sequence. This conclusively proves that the degradation is not inherent to masking; rather, it is caused by PIDNet's imprecise, noisy boundary predictions. By accidentally masking high-gradient static background features adjacent to the dynamic actors, the student model destroyed the critical structural data needed to solve the rotational Perspective-n-Point (PnP) problem.

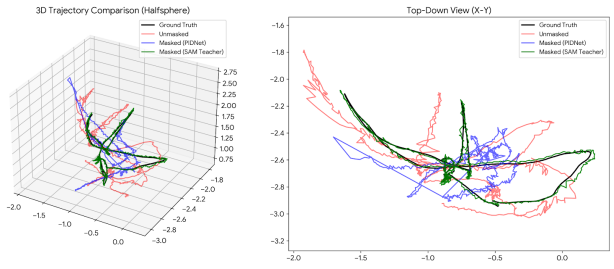


Fig. 4. Trajectory comparison for the `walking_halfsphere` sequence. The noisy boundaries of the PIDNet model (blue) induce feature starvation, resulting in worse performance than the unmasked system (red). Perfect boundary masking via SAM (green) solves the rotational drift.

3) Stationary Camera (`walking_static`): In the `static` sequence, the unmasked baseline naturally performs very well (0.1019m error) because ORB-SLAM's local bundle adjustment easily rejects dynamic outliers when the global geometry remains stationary.

Similar to the rotational sequence, the PIDNet model introduced micro-jitters, increasing the error to 0.2378m due to over-segmentation and the removal of valid static features. Once again, the SAM model validated the core hypothesis, achieving an ATE of 0.0115m. This confirms that if pixel-perfect boundaries can be achieved, dynamic masking is universally beneficial across all kinematic regimes.

VII. DISCUSSION

The results of our evaluation provide critical insights into the current state and future trajectory of dynamic SLAM. By evaluating both a real-time student network (PIDNet) and an offline foundation model (SAM), we are able to separate the fundamental limitations of the semantic masking paradigm from the specific architectural shortcomings of lightweight neural networks.

A. Validation of the Masking Paradigm

Historically, literature on dynamic SLAM has suggested that aggressively masking the center of the camera frame

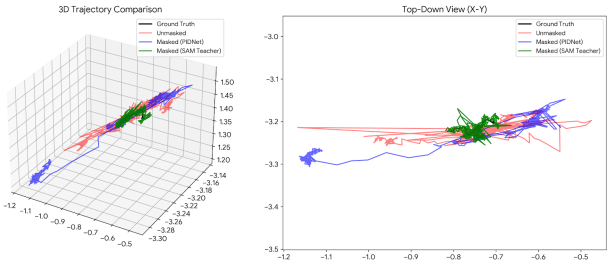


Fig. 5. Trajectory comparison for the `walking_static` sequence. The unmasked baseline (red) natively handles static tracking well. The PIDNet student (blue) introduces micro-jitters due to boundary over-segmentation, whereas the SAM teacher (green) maintains near-perfect stability.

is inherently risky, as it removes potentially useful features and can cause "feature starvation" during complex camera motions. However, our evaluation of the SAM teacher model conclusively disproves this as a fundamental limitation.

Across all three kinematic profiles—pure translation, heavy rotation, and a completely stationary camera—the SAM-masked pipeline achieved near-perfect tracking with Absolute Trajectory Errors ranging from just 1.1 cm to 2.3 cm. This proves that if a SLAM system is provided with pixel-perfect semantic boundaries, feature-based tracking can almost entirely eliminate dynamic drift without degrading the underlying geometric stability of the solver.

B. The Boundary Noise Bottleneck

While the theoretical upper bound of masking is remarkably high, our real-time PIDNet student model exposed the practical limitations of current lightweight segmentation networks. On the purely translational `walking_xyz` sequence, PIDNet successfully rejected enough dynamic outliers to cut the baseline trajectory error in half (from 0.74m to 0.35m).

However, on the `walking_halfsphere` and `walking_static` sequences, the PIDNet student performed significantly worse than doing nothing at all (the unmasked baseline). By comparing this degradation against the SAM baseline, we isolated the root cause: **imprecise boundary segmentation**.

Lightweight networks operating at 40 FPS naturally suffer from inference noise, leading to overly thick or blurred object boundaries. When PIDNet over-segments a human subject, it inadvertently masks out the immediately adjacent static background geometry.

- **In rotational sequences (`halfsphere`):** Features are already rapidly leaving the camera's field of view. Destroying the few remaining high-gradient static features near the center of the frame via a noisy mask crashes

the tracking optimizer, leading to severe structural drift (0.52m error).

- **In stationary sequences (`static`):** The erroneous, frame-by-frame deletion and reappearance of static features due to flickering network boundaries introduces micro-jitters into the PnP solver, causing the camera pose to artificially vibrate (0.23m error).

C. The Real-Time Trade-off

Ultimately, this project highlights the severe latency-accuracy trade-off currently dominating semantic SLAM. The SAM foundation model provides the necessary boundary precision but requires 0.75 seconds per frame, rendering it entirely incompatible with real-time robotic deployment. Conversely, the PIDNet student model perfectly satisfies the 30 FPS strict latency budget (requiring only 0.025 seconds per frame) but lacks the spatial precision required to safely navigate rotational kinematics.

These findings suggest that future research in dynamic SLAM should pivot away from simply identifying *which* pixels are dynamic, and focus heavily on *boundary refinement*. Techniques such as temporal mask smoothing, edge-aware loss functions during student distillation, or utilizing slightly heavier but more precise networks (such as YOLOv8-Seg or FastSAM) may be required to close the performance gap between real-time inference and the foundation model upper bound.

VIII. Future Work

While our experiments validate the theoretical upper bound of semantic masking via the SAM foundation model, they also expose the operational bottlenecks of utilizing lightweight student networks in real-time. Future work must bridge this gap by addressing both the upstream training of the semantic model and the downstream algorithmic integration within the SLAM backend.

A. Advancing the Distillation Pipeline

The primary failure mode of the PIDNet student model was boundary imprecision and frame-to-frame boundary flickering, which introduced micro-jitters into the PnP solver. Future improvements to the training pipeline should focus on mitigating this noise.

First, the spatial distillation process could be enhanced by introducing an edge-aware loss function, heavily penalizing the student model for divergence from the teacher's logits specifically along the high-gradient contours of dynamic subjects. Second, because SLAM operates on sequential data, the student model currently lacks temporal context. Upgrading the distillation pipeline to utilize video-based foundation models (such as SAM 2) and incorporating a temporal consistency loss during training would teach the PIDNet student to output temporally stable masks, drastically reducing the erroneous frame-by-frame deletion and reappearance of static background features.

B. Mask-Aware Feature Allocation

Currently, our integration applies the semantic mask strictly as a post-processing filter. The ORB-SLAM frontend uniformly extracts a fixed quota of FAST corners across the entire image grid, and the dynamic mask subsequently prunes them. Consequently, if a dynamic object occupies 50% of the camera frame, the SLAM system loses 50% of its potential tracking features. This uniform-extraction-then-pruning paradigm is the root cause of the system's vulnerability to feature starvation.

A critical avenue for future work is deep algorithmic integration to achieve *Mask-Aware Feature Allocation*. Instead of generating features blindly, the SLAM frontend should ingest the binary segmentation mask M *prior* to the FAST extraction phase. By analyzing the mask, the algorithm could dynamically re-weight its extraction grid, concentrating the system's total feature quota N exclusively within the static regions. If the static background lacks sufficient natural texture, the system could automatically lower the FAST extraction threshold in those specific unmasked regions to guarantee a robust minimum number of static keypoints.

By ensuring the SLAM backend always receives a fully saturated set of background features, the system could maintain aggressive rotational tracking even when dynamic actors dominate the camera's field of view.

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